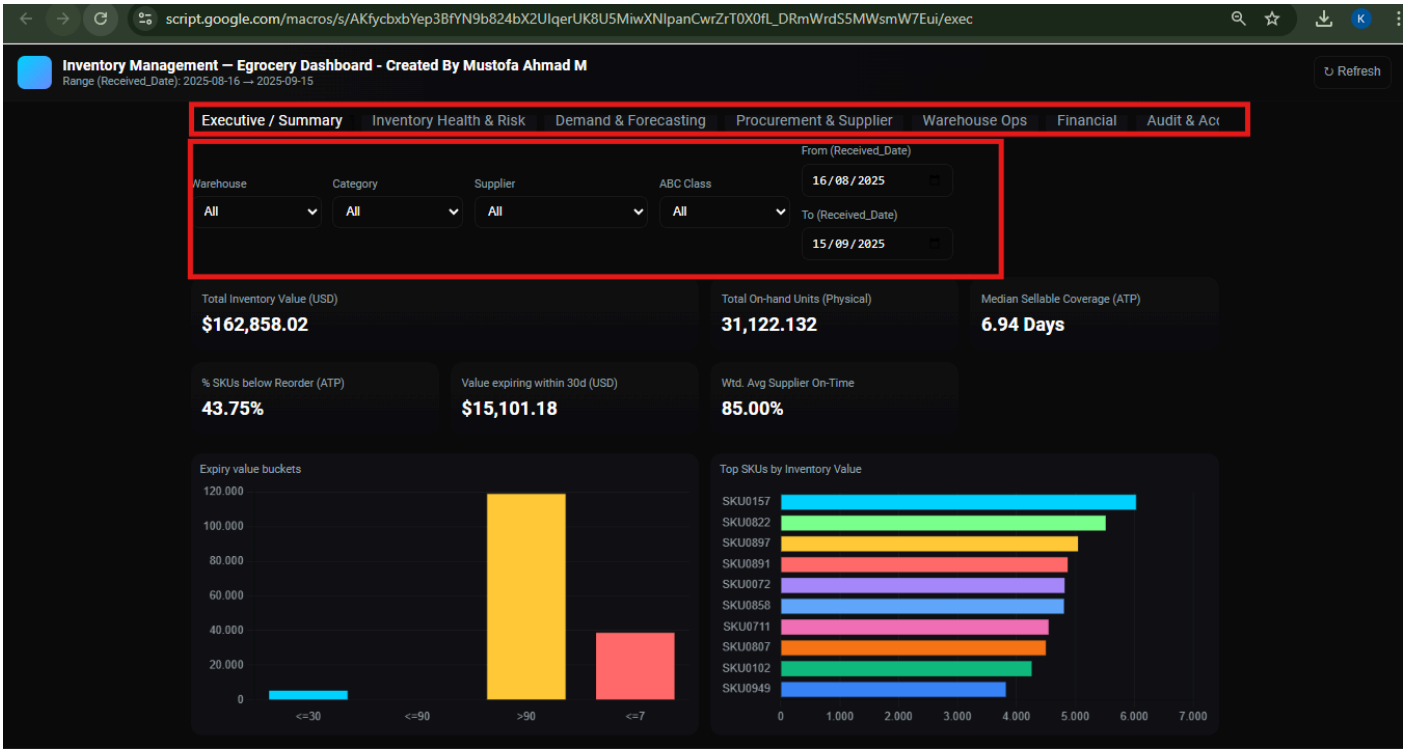
E-Grocery Inventory Management Dashboard

Role Play: Inventory Manager at E-Grocery Company

Subject: Overview and functional guide for the new Inventory Management Dashboard, developed to provide real-time, actionable insights into our e-grocery stock position and performance.

This document outlines the capabilities of our new data-driven Inventory Management Dashboard. As an e-grocery business where freshness and availability are paramount, this tool moves us beyond static reports to a dynamic command center. It is designed to empower us to optimize stock levels, minimize waste from expired goods, improve forecast accuracy, and ensure we have the right products available to meet customer demand, ultimately protecting both our revenue and customer satisfaction.



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1. Data Set Overview [(Click Here To Preview Data Set)]

The dashboard is powered by a central `InventoryData` sheet that captures the complete lifecycle and status of every Stock Keeping Unit (SKU) in our fulfillment centers. Each data field is critical for maintaining a healthy and efficient inventory.

Tabel1

SKU_ID	SKU_Name	Category	ABC_Class	Supplier_ID	Supplier_Name	Warehouse_ID	Warehouse_Location	Batch_ID	Received_Date	Last_Purchase_Da
SKU0001	Pantry Product 13	Pantry	A	S005	PT Agro Raya	WHBDG	Bandung - Rancaekek	BATCH2679	2025-07-14	2025
SKU0002	Fresh Product 112	Fresh Produce	C	S004	PT Nusantara Supplier	WHDPS	Denpasar - Tabanan	BATCH4257	2025-04-08	2025
SKU0003	Meat Product 446	Meat	B	S001	PT Segar Makmur	WHBDG	Bandung - Rancaekek	BATCH6574	2025-03-15	2025
SKU0004	Seafood Product 48	Seafood	A	S007	PT Bakerindo	WHJKT	Jakarta - Cengkareng	BATCH5333	2025-06-10	2025
SKU0005	Personal Product 194	Personal Care	A	S002	PT Indo Fresh	WHDPS	Denpasar - Tabanan	BATCH6925	2025-06-26	2025
SKU0006	Beverages Product 41	Beverages	A	S004	PT Nusantara Supplier	WHJKT	Jakarta - Cengkareng	BATCH8428	2025-06-04	2025
SKU0007	Dairy Product 137	Dairy	C	S002	PT Indo Fresh	WHDPS	Denpasar - Tabanan	BATCH9751	2025-03-31	2025
SKU0008	Meat Product 474	Meat	C	S009	PT Beverage Nusantara	WHBDG	Bandung - Rancaekek	BATCH1916	2025-03-18	2025
SKU0009	Fresh Product 468	Fresh Produce	B	S010	PT DailyDairy	WHSBY	Surabaya - Sidoarjo	BATCH9179	2025-07-17	2025
SKU0010	Meat Product 288	Meat	C	S009	PT Beverage Nusantara	WHSBY	Surabaya - Sidoarjo	BATCH7543	2025-04-13	2025
SKU0011	Fresh Product 25	Fresh Produce	C	S002	PT Indo Fresh	WHBDG	Bandung - Rancaekek	BATCH7916	2025-04-02	2025
SKU0012	Personal Product 271	Personal Care	C	S005	PT Agro Raya	WHDPS	Denpasar - Tabanan	BATCH2876	2025-09-07	2025
SKU0013	Meat Product 81	Meat	C	S008	PT Hygienic Supplies	WHJKT	Jakarta - Cengkareng	BATCH3927	2025-07-04	2025
SKU0014	Pantry Product 312	Pantry	B	S004	PT Nusantara Supplier	WHBDG	Bandung - Rancaekek	BATCH9837	2025-06-06	2025
SKU0015	Fresh Product 476	Fresh Produce	C	S006	PT Frozen Express	WHSBY	Surabaya - Sidoarjo	BATCH4946	2025-07-10	2025
SKU0016	Household Product 390	Household	C	S009	PT Beverage Nusantara	WHBDG	Bandung - Rancaekek	BATCH8787	2025-08-08	2025
SKU0017	Pantry Product 494	Pantry	A	S004	PT Nusantara Supplier	WHDPS	Denpasar - Tabanan	BATCH6107	2025-03-17	2025
SKU0018	Frozen Product 127	Frozen	A	S004	PT Nusantara Supplier	WHJKT	Jakarta - Cengkareng	BATCH4770	2025-06-15	2025
SKU0019	Meat Product 464	Meat	C	S001	PT Segar Makmur	WHSBY	Surabaya - Sidoarjo	BATCH4899	2025-08-22	2025
SKU0020	Beverages Product 452	Beverages	B	S010	PT DailyDairy	WHDPS	Denpasar - Tabanan	BATCH8749	2025-05-11	2025
SKU0021	Seafood Product 211	Seafood	B	S008	PT Hygienic Supplies	WHJKT	Jakarta - Cengkareng	BATCH1993	2025-03-21	2025
SKU0022	Dairy Product 230	Dairy	C	S003	PT Prima Distribusi	WHMDN	Medan - Deli Serdang	BATCH8579	2025-07-25	2025
SKU0023	Household Product 51	Household	B	S001	PT Segar Makmur	WHDPS	Denpasar - Tabanan	BATCH4872	2025-09-06	2025

Core Product & Location Identifiers

- `SKU_ID` / `SKU_Name`: Unique identifiers for each product.
- `Warehouse_ID`: The fulfillment center where the stock is located.
- `Category`: The product category (e.g., Fresh Produce, Dairy, Frozen Goods).
- `Supplier_Name`: The vendor who supplies the product.
- `ABC_Class`: The inventory categorization (A, B, C) based on value and sales velocity, crucial for prioritization.

Stock Quantity & Status

- `Quantity_On_Hand (QOH)`: The total physical quantity of an item in the warehouse.
- `Quantity_Reserved`: Stock allocated to processed but not-yet-shipped customer orders.
- `Quantity_Committed`: Stock allocated for promotions, transfers, or other internal uses.
- `Received_Date`: The date stock was last received into the warehouse.
- `Expiry_Date`: The expiration date of the batch, a critical field for perishable goods.

Planning & Velocity Metrics

- `Avg_Daily_Sales`: The average number of units sold per day, used for demand forecasting.
- `Reorder_Point`: The stock level at which a new purchase order should be triggered.
- `Safety_Stock`: The buffer stock level maintained to mitigate stockouts from unexpected demand or supplier delays.
- `Lead_Time_Days`: The average time in days from placing a purchase order to receiving the goods.

Financial & Quality Metrics

- `Total_Inventory_Value_USD`: The total financial value of the on-hand stock for an SKU.
- `Stock_Age_Days`: The number of days the current stock has been held.
- `Damaged_Qty` / `Returns_Qty`: The quantity of units recorded as damaged or returned.
- `Supplier_OnTime_Pct`: The percentage of time a supplier delivers within the agreed window.
- `Demand_Forecast_Accuracy_Pct`: The accuracy percentage of our sales forecast versus actual sales.
- `Audit_Variance_Pct`: The percentage variance found during physical cycle counts versus system records.

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2. Key Metrics & Measurements

The dashboard's intelligence lies in its calculated metrics, which are organized into tabs for focused analysis. This dashboard utilizes a sophisticated **Available-to-Promise (ATP)** logic (`ATP = QOH - Reserved - Committed`) for most of its sellable stock calculations, ensuring we only analyze inventory that is truly available for sale.

A. Inventory Health & Risk

- **% SKUs Below Safety Stock (using ATP):**
  - **Objective:** To identify the percentage of items that have breached their minimum safety buffer, based on sellable stock (ATP).
  - **Goal:** To keep this number as low as possible, ideally **below 5%**, to minimize stockout risk and ensure a consistent customer experience.
- **Value Expiring within 30 Days (USD):**
  - **Objective:** To quantify the financial risk associated with products nearing their expiration date.
  - **Goal:** To proactively reduce this value through targeted promotions, markdowns, or flash sales, minimizing financial losses from spoilage.
- **Dead Stock Value (USD):**
  - **Objective:** To identify the value of stock that has not moved for an extended period (e.g., >180 days).
  - **Goal:** To clear dead stock through liquidation or bundling strategies, freeing up capital and warehouse space.

B. Demand & Forecasting

- **Weighted Avg. Forecast Accuracy:**
  - **Objective:** To measure the accuracy of our demand planning, weighted by inventory value to prioritize high-value items.
  - **Goal:** To achieve a weighted forecast accuracy of **over 85%**. Higher accuracy leads to better purchasing decisions, reducing both overstock and understock situations.
- **Median Sellable Coverage (Days):**

    - **Objective:** To determine the median number of days we can continue selling our products based on the current sellable stock (ATP) and average daily sales.
    - **Goal:** To maintain a healthy coverage range (e.g., 7-14 days depending on the category) to balance availability with the risk of spoilage.

C. Procurement & Supplier Performance

- **Weighted Avg. Supplier On-Time %:**
    - **Objective:** To measure the reliability of our suppliers' delivery schedules, weighted by the value of goods they provide.
    - **Goal:** To ensure our strategic suppliers maintain an on-time delivery rate of **over 95%**, as delays directly impact our own stock availability.
- **Average Lead Time (Days):**
    - **Objective:** To monitor the average time it takes to replenish stock from suppliers.
    - **Goal:** To work with suppliers to reduce lead times, allowing us to hold less safety stock and be more agile in our inventory planning.

D. Financial Metrics

- **ABC Value Distribution:**
    - **Objective:** To visualize the breakdown of total inventory value across A, B, and C class items.
    - **Goal:** To confirm that our management focus aligns with the 80/20 principle, where the majority of our capital is tied up in the "A" class items that drive the most value.

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3. Key Performance Indicators (KPIs)

For reporting and strategic management, we will focus on the following core KPIs, which provide the most accurate snapshot of our inventory health and performance.

- **KPI 1: Median Sellable Coverage (based on ATP)**
    - **Definition:** The median number of days we can sustain sales with our current Available-to-Promise inventory.
    - **Importance:** This is the primary health metric for an e-grocery business. It balances the risk of stockouts against the risk of spoilage. A coverage that is too high indicates overstocking and waste risk; too low indicates a risk to sales.
    - **Strategic Goal: Maintain median coverage between 7 and 14 days for ambient goods, and between 3 and 5 days for fresh categories.**
- **KPI 2: Value Expiring within 30 Days (USD)**
    - **Definition:** The total financial value of all inventory set to expire in the next 30 days.
    - **Importance:** This KPI represents the most immediate and tangible financial risk to the business. It is a direct measure of potential waste and lost profit.
    - **Strategic Goal: Keep this value below 2% of the total inventory value through proactive inventory management and targeted sales initiatives.**

- **KPI 3: Weighted Average Demand Forecast Accuracy**
    - **Definition:** The value-weighted accuracy of our sales forecasts.
    - **Importance:** Accurate forecasting is the foundation of efficient inventory management. This KPI dictates how effectively we can order from suppliers, minimizing carrying costs from overstock and lost sales from stockouts.
    - **Strategic Goal: Achieve a sustained weighted forecast accuracy of ≥ 85% to optimize our purchasing and capital allocation.**
- **KPI 4: % SKUs Oversold**
    - **Definition:** The percentage of SKUs where committed and reserved quantities exceed the physical on-hand stock.
    - **Importance:** This is a critical data integrity and customer experience KPI. An oversold item guarantees a failed order, leading to customer dissatisfaction and operational chaos.
    - **Strategic Goal: Maintain a target of 0%. Any instance of an oversold SKU must trigger an immediate root cause analysis.**

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4. Dashboard Functionality Guide [(Click Here To Preview Dashboard)]

The dashboard is designed for ease of use, providing powerful analytical capabilities through a simple interface.

A. Header & Filters

- **Header:** Displays the dashboard title and the active date range being analyzed.
- **Filters:** A persistent top-row allows you to slice the data by **Warehouse**, **Category**, **Supplier**, and **ABC Class**. A **Date Range** filter (based on `Received_Date`) is also available for time-based analysis.
- **Refresh Button:** Applies any changes to the filters and reloads the dashboard with the relevant data.

B. Navigation Tabs The core of the dashboard is organized into logical tabs, each providing a different lens through which to view inventory performance.

- **Executive / Summary:** A high-level overview for senior management, showing top-line metrics like total inventory value and overall stock coverage.
- **Inventory Health & Risk:** Focuses on risk metrics such as SKUs below safety stock, dead stock value, and the percentage of oversold items. This is the tab for identifying immediate problems.
- **Demand & Forecasting:** Provides insights into forecast accuracy and inventory turnover, helping to validate demand planning.
- **Procurement & Supplier:** Centers on supplier performance, showing weighted on-time delivery rates and average lead times.
- **Warehouse Ops:** A view tailored for warehouse-specific analysis, focusing on physical unit counts and values within the filtered warehouse.

- **Financial:** Contains all value-based metrics, including total inventory value, expiring stock value, and the ABC value distribution chart.
- **Audit & Accuracy:** Displays metrics related to inventory record accuracy, such as audit variance and oversold flags.
- **Alerts / Action Center:** The most action-oriented tab, highlighting SKUs that are below reorder points, below safety stock, and have high expiring value, flagging them for immediate attention.

C. KPI Cards & Charts Each tab contains a dynamic layout of **KPI Cards** for at-a-glance metrics and **Charts/Tables** for deeper analysis. For example:

- The **Financial** tab features a pie chart visualizing the **ABC Value Distribution**, showing where capital is concentrated.
- The **Health & Risk** tab includes a table of the **Top 10 SKUs by Stock Age** to quickly identify slow-moving items.
- The **Executive** tab shows a bar chart of the **Top SKUs by Inventory Value**, providing a clear view of the most significant products.